NLP Homework 3

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1 Methodology

1.1 Base Model

We use the Qwen 3-4B model as our base architecture. Qwen is a series of large language models developed by Alibaba Cloud, known for strong performance on Chinese language tasks. The 4B parameter variant provides a good balance between model capability and computational efficiency, making it suitable for fine-tuning on consumer-grade hardware.

1.2 Dataset

Huatuo26M-Lite The training data comes from FreedomIntelligence/Huatuo26M-Lite, a curated Chinese medical dataset. Each sample consists of a medical question and a detailed answer. For example, one sample asks:

"什么是医学伦理学，它在医疗领域有何重要性？"

(Translation: "What is medical ethics, and what is its importance in the medical field?")

The dataset covers diverse medical topics including medical ethics, clinical practice, disease diagnosis, treatment protocols, and medical terminology.

1.3 Parameter-Efficient Fine-Tuning with LoRA

LoRA Concept Low-Rank Adaptation (LoRA) works by injecting trainable low-rank decomposition matrices into each layer of the transformer architecture. Instead of updating the full weight matrix W, LoRA represents the weight update as $\Delta W = BA$, where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with rank $r \ll \min(d, k)$.

2 Experimental Setup

2.1 Training Hyperparameters

We configure the training process with the following hyperparameters, optimized for efficient training on limited GPU memory:

Parameter	Value
Number of Epochs	1
Batch Size (Train)	8
Batch Size (Eval)	8
Gradient Accumulation Steps	4
Effective Batch Size	32 (8×4)
Learning Rate	2e-7
Weight Decay	0.001
Warmup Ratio	0.03
LR Scheduler	Cosine
Gradient Checkpointing	True

Table 1: Training hyperparameters configuration

2.2 Generation Configuration

For model evaluation, we implement a custom generation function with the following characteristics:

- **Template**: Uses Phoenix conversation format with role-based prompting
- **Sampling Strategy**: Greedy decoding (do_sample=False) for deterministic outputs
- **Max New Tokens**: 64 tokens maximum for generated responses
- **Repetition Penalty**: 1.0 (no penalty) to allow natural repetition of medical terms

3 Results

3.1 Pre-Training Model Behavior

Before fine-tuning on the medical dataset, we tested the base Qwen 3-4B model on two queries to establish baseline behavior:

Query 1: "What's the weather like today?"

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Response: I'm sorry, but I can't provide real-time information
like today's weather. To find out the current weather, I
recommend checking a reliable weather website or app such as
Weather.com, AccuWeather, or your smartphone's built-in
weather feature.
```

This response demonstrates the base model's awareness of its limitations regarding real-time information, showing appropriate task understanding.

Query 2: "Who are you?"

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Response: I'm Qwen, a large-scale language model developed by
Alibaba Cloud's Tongyi Lab. You can call me Qwen. I'm designed
to assist with a wide range of tasks, such as answering
questions, writing stories, creating documents, coding, and more.
```

The model correctly identifies itself and describes its general capabilities, indicating proper self-awareness and instruction-following ability.

3.2 Dataset Sample Analysis

The training data consists of high-quality medical question-answer pairs. A representative example regarding medical ethics:

Question (Chinese): 什么是医学伦理学，它在医疗领域有何重要性?

Answer (Chinese): 医学伦理学是研究医疗领域伦理问题的学科。它涉及研究医疗专业人员、患者和其他相关利益相关者之间的伦理关系，以及在医疗实践中出现的道德困境。

医学伦理学在医疗领域具有以下重要性:

1. 保护患者权益：医学伦理学关注患者的权益和尊严。它确保医疗决策是以患者的最大利益为出发点，并尊重患者的自主权和知情同意权。

2. 促进医务人员职业道德：医学伦理学提供了医务人员在面对道德困境时的指导原则，帮助他们保持专业的道德标准和行为规范。

3. 增加医疗决策的公正性：医学伦理学关注公正和公平的医疗分配原则。它确保资源在医疗领域的分配是公正和可持续的。

4. 促进研究伦理：医学伦理学对医学研究进行伦理审查，确保研究参与者的权益和福利得到保护，并确保研究过程是符合伦理标准的。

5. 保护医疗机构声誉：医学伦理学的遵循有助于确保医疗机构遵守伦理原则，保护其声誉和公众信任度。

总之，医学伦理学在医疗领域的重要性在于维护患者权益、指导医务人员的职业道德行为、促进医疗决策公正和保护医学研究伦理。它为医疗行业提供了一个道德框架，确保医疗服务的质量和道德高于一切。”

The answer comprehensively covers patient rights, professional ethics, fairness in medical decisions, research ethics, and institutional reputation protection.

This example demonstrates the dataset's depth and structure, with detailed explanations suitable for training a medical-domain language model.

4 Conclusion

This project successfully demonstrates the application of LoRA for parameter-efficient fine-tuning of the Qwen 3-4B model on Chinese medical data. The methodology combines efficient training techniques including gradient accumulation, gradient checkpointing, and low-rank adaptation to make large-model fine-tuning accessible on limited hardware.

The work lays foundation for developing specialized medical AI assistants in Chinese, with potential applications in medical education, clinical decision support, and patient communication. Future iterations should focus on comprehensive evaluation, extended training, and deployment optimization for real-world medical applications.